

# V11.9 Final - Spin as Lag-Field Holonomy

Compton Circulation, SU(2) Closure, Magnetic Moment Interface, and Vertex Dressing

Status: final V11.9 sector note

Branch: V11 particle-ontology / lag-field geometry

Support toys: V11.9A through V11.9I

Repro policy: scripts-only repro pack; generated outputs are regenerated locally

Guardrail: representation and interface sector; not a derivation of spin-statistics, Pauli exclusion, or the full anomalous magnetic moment

## 0. Abstract

V11.9 opens and develops the SFT spin sector. The core proposal is:

Spin is lag-field holonomy of Compton circulation.

In SFT, a massive elementary particle is treated as a maintained geometric process: a localized Compton oscillator with internal phase circulation and self-consistent lag-field accounting. Spin is not ordinary rotation of a tiny classical object. Spin is the closure topology and handedness of the particle's internal phase frame.

The sector was tested through V11.9A-I: closure topology, SU(2) handedness, Stern-Gerlach projection, Compton spin-scale alignment, magnetic-moment ring-current audit, anomaly-scale budget, Pauli/Dirac coupling, winding-output structure, and vertex dressing.

Final claim level:

SFT can represent spin as internal lag-field holonomy with correct SU(2) closure.

The spinor-aligned Compton scale naturally lands on  $\hbar / 2$ .

Leading magnetic moment  $g = 2$  is recovered at the Pauli/Dirac interface.

The anomaly scale  $\alpha / (2 \pi)$  emerges naturally from normalized Compton-lag vertex overlap.

The measured anomalous magnetic moment is not derived.

## 1. Established Physics Guardrails

Spin is not ordinary rotation of a small object. A spin-1/2 particle is not a tiny sphere spinning around an axis. The required structure is spinorial.

Ordinary spatial rotations are described by  $SO(3)$ . Spinors live in  $SU(2)$ , the double cover of  $SO(3)$ . Therefore:

$$2 \pi: \psi \rightarrow -\psi$$

$$4 \pi: \psi \rightarrow \psi$$

Any SFT spin model must respect this. The correct SFT translation is not little-ball rotation. It is topological closure of internal phase circulation.

## 2. SFT Translation

SFT translates rest mass into maintained Compton ticking:

$$f_C = m c^2 / h$$

$$\omega_C = m c^2 / \hbar$$

V11.9 adds the missing topological degree of freedom:

$$\text{mass} = \text{how fast the maintained clock ticks}$$

$$\text{spin} = \text{how the internal phase frame closes under circulation}$$

## 3. V11.9A - Closure and Lag-Holonomy

V11.9A tested simple closure versus spinor closure.

$$\text{simple loop: } U(\theta + 2\pi) = U(\theta)$$

$$\text{spinor loop: } U(\theta + 2\pi) = -U(\theta)$$

$$\text{spinor loop: } U(\theta + 4\pi) = U(\theta)$$

V11.9A showed that SFT can encode simple  $2\pi$  closure and spinor  $4\pi$  closure as distinct internal lag-holonomy classes. The first handedness check exposed a diagnostic issue: endpoint phase alone cannot distinguish  $+\pi$  from  $-\pi$  because both map to  $-1$ . This was fixed in V11.9B.

Earned: SFT can represent simple and spinor closure topologies.

## 4. V11.9B - $SU(2)$ Closure and Handedness

V11.9B upgraded the toy to two-component  $SU(2)$  spinors.

$$R_n(\theta) = \cos(\theta/2) I - i \sin(\theta/2) (\mathbf{n} \cdot \boldsymbol{\sigma})$$

$$R(2\pi) = -I$$

$$R(4\pi) = I$$

For ordinary SO(3) vector rotations, closure occurs after 2 pi. Thus the SU(2)/SO(3) double-cover distinction is correctly represented.

Handedness was measured by unwrapped phase slope:

$$\text{spin-up } z \rightarrow \exp(-i\theta/2) \text{ spin-up } z ; \text{ slope} = -0.5$$

$$\text{spin-down } z \rightarrow \exp(+i\theta/2) \text{ spin-down } z ; \text{ slope} = +0.5$$

Earned: handedness is path slope, not wrapped endpoint phase.

## 5. V11.9C - Stern-Gerlach Projection Geometry

V11.9C tested whether the SU(2) spinor-holonomy representation supports the standard spin-projection rule.

$$P(+ \text{ along } \mathbf{m} \mid + \text{ along } \mathbf{n}) = \cos^2(\gamma/2) = (1 + \mathbf{n} \cdot \mathbf{m})/2$$

The toy passed to numerical roundoff. It also confirmed sequential Stern-Gerlach reset:

$$z \rightarrow x \rightarrow z$$

$$P(z+ \rightarrow x+ \rightarrow z+) = 1/4$$

$$P(z+ \rightarrow x+ \rightarrow z-) = 1/4$$

Spin-axis projectors were noncommutative, as expected.

Earned: spin-up/down are projections of internal spinor holonomy relative to a chosen axis.

## 6. V11.9D - Compton Spin-Scale Alignment

V11.9D tested whether c-speed Compton/lag-field circulation can land on the spin scale  $\hbar/2$ .

$$\omega_{\text{int}} = \kappa \omega_{\text{C}}$$

$$r = c / \omega_{\text{int}} = \hbar / (\kappa m c)$$

$$L = m c r = \hbar / \kappa$$

$$\kappa = 1 \rightarrow L = \hbar$$

$$\kappa = 2 \rightarrow L = \hbar / 2$$

For a spinor requiring 4 pi closure:

$$T_{\text{spinor}_4\pi} / T_{\text{C}} = 2 / \kappa$$

$$\text{at } \kappa = 2: T_{\text{spinor}_4\pi} = T_{\text{C}}$$

Earned: one Compton period equals one full 4 pi spinor closure at  $L = \hbar / 2$ .

## 7. V11.9E - Magnetic Moment Counting Audit

V11.9E tested the proposed motion explanation that two charge laps per one spinor mechanical closure might by itself produce leading  $g = 2$ . It does not.

The ordinary current formula already counts laps per second. If charge and spin share the same loop radius, literal two-lap counting is already included in the circulation frequency.

$$\text{same-loop ring current gives } g = 1$$

V11.9E found conditional ways to represent  $g = 2$ , including an independent topological current factor or an effective charge-current radius distinct from the spin/inertia radius. These require additional justification.

Earned: naive ring-current motion does not derive leading  $g = 2$ .

## 8. V11.9G - Pauli/Dirac Coupling Interface

V11.9G tested the correct leading magnetic-moment mechanism: spinor covariant electromagnetic coupling.

$$(\boldsymbol{\sigma} \cdot \boldsymbol{\pi})^2 = \pi^2 - q \hbar \boldsymbol{\sigma} \cdot \mathbf{B}$$

$$\pi = p - q A$$

The Pauli Hamiltonian contains:

$$H = \pi^2 / (2m) - (q \hbar / 2m) \sigma \cdot B$$

Using  $S = (\hbar/2) \sigma$  and comparing to  $H_Z = -g (q/2m) S \cdot B$  gives:

$$g = 2$$

V11.9G passed analytically and in the finite operator test. The best-fit Zeeman coefficient from the spinor Hamiltonian was  $g = 2.0$ .

Earned: leading  $g = 2$  is closed at the Pauli/Dirac spinor-covariant interface.

This is not a derivation of the anomalous magnetic moment.

## 9. V11.9F and V11.9I - Anomaly Scale and Vertex Dressing

The anomalous magnetic moment is:

$$a = (g - 2) / 2$$

V11.9F showed that the natural scale  $\alpha/(2 \pi)$  lands in the correct anomaly neighborhood. For the electron, the required coefficient was:

$$a = C \alpha/(2 \pi), C = 0.998487$$

V11.9I then tested an SFT-native mechanism:

$$a_{\text{SFT}} = \alpha/(2 \pi) C_{\text{lag}}$$

For normalized spinor-bilinear Compton-lag vertex overlap, the toy found:

$$C_{\text{lag}} = 1$$

A mild retarded phase-sensitive correction could match the electron-scale coefficient with small parameters:

$$\tau = 2.820069, \epsilon = -0.013522$$

Earned: normalized Compton-lag vertex overlap naturally yields the  $\alpha/(2 \pi)$  anomaly scale.

Guardrail: the measured anomalous magnetic moment is not derived.

## 10. V11.9H - Winding Pattern and Field Output

V11.9H formalized the generator-intuition bridge:

motion + winding topology + coupling geometry -> field-output fingerprint

The toy showed that spinor sign flips can be invisible to bilinear standalone field output; handedness appears as phase-current direction; different winding/current profiles produce distinct Fourier/multipole fingerprints; and zero-mean AC winding can have no DC dipole but nonzero rectified mean-square output.

This is not a claim that particles are literal generators. It is a structural analogy pass.

Earned: field outputs remember internal winding geometry.

## 11. What V11.9 Establishes

1. SFT can represent spin as internal lag-field holonomy.
2. Simple/integer loops close after  $2\pi$ .
3. Spinor/half-twist loops flip after  $2\pi$  and close after  $4\pi$ .
4.  $SU(2)$  spinor closure is correctly represented.
5.  $SO(3)$  vector closure and  $SU(2)$  spinor closure are correctly distinguished.
6. Handedness is encoded in unwrapped phase slope, not wrapped endpoint phase.
7.  $\Psi$  and  $-\Psi$  give identical standalone measurement probabilities.
8. The spinor sign flip is visible in relative/interferometric phase.
9. The Stern-Gerlach projection rule is represented correctly.
10. Sequential measurement reset is represented correctly.
11. Spin-axis projectors are noncommutative as expected.
12. Compton/spinor scale alignment naturally gives  $\hbar/2$  at  $\kappa = 2$ .
13. Naive ring-current motion gives  $g = 1$ , not  $g = 2$ .
14. Leading  $g = 2$  is recovered at the Pauli/Dirac spinor-covariant interface.
15. The anomaly scale  $\alpha/(2\pi)$  is naturally present.
16. Normalized Compton-lag vertex overlap gives  $C_{\text{lag}} = 1$  at the mechanism-scale level.
17. Winding patterns produce distinct output fingerprints.
18. The exchange-sign seed is present but not yet a spin-statistics derivation.

## 12. What V11.9 Does Not Establish

1. A first-principles derivation of electron spin from SFT lag geometry.
2. A derivation of the spin-statistics theorem.

3. A derivation of Pauli exclusion.
4. A complete derivation of the electron magnetic moment including the anomaly.
5. A derivation of  $g = 2$  from naive ring-current motion.
6. A full replacement for SU(2) spinor quantum mechanics.
7. A full replacement for QED.
8. A derivation of the Born rule or measurement collapse.
9. A complete classification of all particle spins.
10. A precision calculation of  $F_2(0)$ .

## 13. Open Problems

### 13.1 Dynamical origin of half-twist holonomy

Can a self-generated lag connection naturally produce half-twist holonomy for fermions?

### 13.2 Spin-statistics bridge

Can spinor lag-field holonomy reproduce exchange antisymmetry in many-particle configuration space?

### 13.3 Pauli exclusion

Can exclusion be interpreted as geometric incompatibility of identical spinor-holonomy circulation states?

### 13.4 Pauli/Dirac interface from SFT dynamics

V11.9G closes leading  $g = 2$  at the interface level. A future note must ask whether SFT lag geometry can dynamically produce that interface rather than importing it.

### 13.5 Anomalous magnetic moment

V11.9I gives the natural anomaly scale, but the measured anomaly remains a precision QED observable unless SFT reproduces the full vertex form-factor structure or predicts a residual below experimental bounds.

### 13.6 Coupling to V11.3 Higgs Contact

Mass sets the clock rate. Spin sets closure topology. The Higgs-contact sector and spin-holonomy sector should eventually be combined into one particle ledger.

### 13.7 Coupling to V11.2 Dent Field

If particles are localized defects in their own lag geometry, spinor holonomy should affect interference, exchange, and measurement behavior. This belongs in the dent-field/spin interface.

## 14. Final Synthesis

The mature V11.9 picture is not particles as tiny spinning balls.

particles are maintained oscillatory defects whose internal phase frames carry  
quantized holonomy

Mass sets the Compton ticking rate. Spin sets the closure topology. Measurement reads projections of the holonomy relative to an axis. The spinor-aligned Compton scale lands on  $\hbar/2$ . Leading  $g = 2$  comes from the Pauli/Dirac spinor-covariant interface, not from naive current-loop motion. The anomaly scale emerges naturally from normalized Compton-lag vertex overlap, but the measured anomaly remains underived.

Spin is lag-field holonomy of Compton circulation.

## 15. One-Line Summary

V11.9 proposes that spin is not ordinary rotation but the  $SU(2)$  closure topology of a particle's internal Compton/lag-field circulation: simple loops close after  $2\pi$ , spinor loops flip after  $2\pi$  and close after  $4\pi$ , projection geometry is represented correctly, the spinor-aligned Compton scale lands on  $\hbar/2$ , leading  $g = 2$  is recovered at the Pauli/Dirac interface, and normalized Compton-lag vertex overlap naturally yields the  $\alpha/(2\pi)$  anomaly scale while the measured anomaly remains open.